4.2 Power Series

In the last section we looked *n*th Maclaurin and Taylor polynomials which were used to approximate a function *f* about a point $x = x_0$. It seemed reasonable that as *n* increased $p_n(x)$ would provide better and better approximations, and may even possible converge to $f(x)$ as $n \to \infty$.

Definition:

Maclaurin and Taylor series differ form the series that we considered in 3.2 – 3.8 in that their terms are not merely constants, but also involve variables. These are examples of what we call a *power series*.

Definition:

Example 1: Consider the power series $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots$.

Definition:

Note: As is indicated by the above definition for power series, we adopt the convention that $\left(x-x_0\right)^0=1$ *even when* $x=x_0$.

Example 2: For what values of x is the series $\sum_{n=0}^{\infty} n!x^n$ convergent?

Example 3 (you try): For what values of x does the series $\sum_{n=1}^{\infty} \frac{\left(x-3\right)^n}{n}$ converge?

Theorem: For a given power series $\sum_{n=0}^{\infty} c_n \left(x - x_0\right)^n$ there are only three possibilities:

Proof: Beyond the scope of the course.

Definition:

Definition:

Example 4: Consider the series (from the previous example): $\sum_{n=0}^{\infty} n!x^n$.

Example 5: Consider the series (from the previous example): $\sum_{n=1}^{\infty} \frac{\left(x-3\right)^n}{n}$.

Example 6 (you try): Find (with justification) the radius of convergence and interval of convergence of the series: $\sum_{n=1}^{\infty}\left(\frac{(2x-1)^n}{5^n\sqrt{n}}\right)$.